mislay physical devices. While biometric factors can be appealing, they often have high error rates. For this reason, the most secure authentication systems require two or more different factors. For example, your bank may require you to enter a password and then use a device with your bank card to generate a unique login code. This is known as *two-factor authentication* (2FA) or *multi-factor authentication* (MFA).

> **DEFINITION** ***Two-factor authentication (2FA) or multi-factor authentication (MFA) require a user to authenticate with two or more different factors so that a compromise of any one factor is not enough to grant access to a system.***

Note that an authentication factor is different from a credential. Authenticating with two different passwords would still be considered a single factor, as they are both based on something you know. On the other hand, authenticating with a password and a time-based code generated by an app on your phone counts as 2FA because the app on your phone is something you have. Without the app (and the secret key stored inside it), you would not be able to generate the codes.

1.5.2 Access control and authorization

In order to preserve confidentiality and integrity of your assets, it is usually necessary to control who has access to what and what actions they are allowed to perform. For example, a messaging API may want to enforce that users